

Exploring the Quality of TikTok-Based Diabetes Self-Management Education in English and Spanish: A Digital Health Study

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Abstract

Purpose: The purpose of this study was to evaluate TikTok videos about type 2 diabetes mellitus (T2DM) in English and Spanish, with a focus on the Association of Diabetes Care and Education Specialists' 7 self-care behaviors (ADCES7).

Methods: This descriptive study analyzed 300 TikTok videos via English and Spanish hashtags related to T2DM and categorized self-management education content utilizing the ADCES7 self-care behavior framework. Video content creators were categorized into 3 groups: health care professionals, personal accounts, and companies. The Global Quality Scale (GQS) was used to assess video quality from 1 (poor) to 5 (excellent). User engagement metrics were recorded to examine differences across videos.

Results: The analysis of videos revealed that healthy eating was the most frequently addressed ADCES7 self-care behavior, followed by reducing risk. In contrast, problem-solving and healthy coping were the least represented. Spanish-language videos emphasized healthy coping more than English content. Health care professionals' videos had higher GQS scores than personal accounts and companies. However, overall GQS scores remained low—engagement varied by source, with personal creators generating the highest levels of likes and comments.

Conclusions: Traditional in-person diabetes education has limited accessibility, especially for various groups, including Hispanic communities. This study found that TikTok content often lacked high-quality and comprehensive coverage of ADCES7 self-care behaviors, with personal accounts generating the most engagement despite lower quality. Given TikTok's rapid growth, there is an urgent need to improve the quality of diabetes-management content to ensure users receive accurate and reliable information.

By the year 2050, it is projected that 33% of adults in the United States will develop type 2 diabetes mellitus (T2DM),¹ and this chronic disease is expected to affect Hispanic adults at an even higher rate, with estimates suggesting that up to 50% may be impacted.² Hispanic adults face more significant challenges with diabetes-related complications and experience higher mortality rates compared to the overall US population.³ For individuals living with T2DM, diabetes self-management education and support (DSMES) represents a crucial intervention that facilitates healthy lifestyle modifications and promotes optimal disease management. DSMES offers essential health benefits, including improved adherence to treatment regimens, better blood glucose control, increased self-efficacy, reduced diabetes-related complications, and an enhanced quality of life.⁴ However, participation rates in DSMES programs are notably low, with fewer than 5% of eligible individuals taking advantage of these resources.⁵ Moreover, the utilization of DSMES services is even lower among Hispanic populations and other underserved minority groups compared to non-Hispanic White Americans.⁶ Lower participation in DSMES programs is frequently related to financial barriers, lack of insurance, program accessibility, lower socioeconomic status, culture, language, and timing of class availability.⁵

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Social Media for Diabetes Information, Education, and Support

Traditional in-person DSMES programs continue to benefit those who can enroll and fully participate; however, face-to-face classes have limited reach.⁷ Current and future generations raised in a digital ecosystem are poised to rely increasingly on social media as a primary health information and support source.⁸ The growing use of digital health technologies, such as smartphone applications (apps), wearables, sensors, and digital health communities, may allow for developing tailored behavior change interventions.⁹ Consumers actively seek diabetes information through social media,¹⁰ and they tend to trust the recommendations of social media influencers.¹¹ Several studies highlight a growing trend among individuals with T2DM who are turning to social media for diabetes information, education, and support.^{8,10,12-14}

Although the value of DSMES delivered through social media platforms is a new area of research with limited studies, a recent investigation suggests that social media may serve as a reliable source of health information for Spanish-speaking patients with chronic diseases.¹³ Emerging technologies and digital platforms provide opportunities for DSMES in accessible and engaging ways. For example, social media-based microlearning (SMBM) offers a unique approach by creating smaller diabetes-focused education modules.⁷ SMBM's dynamic learning environment integrates audio, music, short videos, photos, and infographics and has demonstrated positive improvements in self-efficacy, self-care, and diabetes knowledge.^{7,14}

Hispanic Adults and Social Media Use

Short-video social media platforms, such as TikTok, are experiencing rapid growth due to their innovative formats and naturally engaging content. These platforms facilitate higher levels of interaction through short videos that feel more authentic, promoting closer and more informal connections between viewers and content creators.¹⁵

Social media platforms can be a practical and effective tool for connecting with individuals who have T2DM, particularly within the Hispanic community, by empowering them through self-management education. With 98% of Hispanics in the United States owning a cellphone and having one of the highest rates of smartphone dependency for internet access, particularly among lower-income households and those with less formal education,¹⁶ social media has the potential to foster engagement and create supportive communities that encourage healthier lifestyles. Notably, TikTok, the fastest-growing short video platform,¹⁷ is particularly popular among Hispanic adults in the United States, with 49% of them engaging with the app, surpassing engagement rates among Black (39%), Asian (29%), and White adults (28%).¹⁸ For many Hispanic adults with T2DM, diabetes education through social media or digital storytelling may be particularly effective, supported by the strong oral traditions in Latin cultures.³

Diabetes and Social Media

The proliferation of social media has changed how people consume information. The rapid increase and widespread use have also led to the dissemination of poor-quality information and the promotion of treatments that deviate from standards of care, such as videos that discourage prescribed insulin use, potentially leading to harm. Misinformation about diabetes is a long-standing issue, exemplified during the COVID-19 pandemic,¹⁹ when anti-vaccination groups falsely claimed vaccines cause type 1 diabetes.²⁰ Previous research has evaluated the usefulness of T2DM videos available on YouTube, focusing on integrating the Association of Diabetes Care and Education Specialists' 7 self-care behaviors (ADCES7).²¹ They found a less than 50% chance of finding videos covering ADCES7 topics and a high risk of encountering misleading information.²² Previous studies examining the quality and completeness of diabetes-related videos posted on TikTok, BiliBili, Weibo, and WeChat²³⁻²⁵ emphasized the need to assess the quality of videos across diverse languages and the importance of rigorous evaluation to determine the quality and reliability of health information posted on these social media platforms. The present landscape reveals a gap in the assessment of the quality of content pertaining to T2DM available in both English and Spanish on the TikTok platform. Consequently, the purpose of this study was to evaluate TikTok videos about T2DM in both English and Spanish, focusing on the ADCES7 framework.

Research Questions

Addressing this gap is crucial for understanding how TikTok content aligns with ADCES7 and for evaluating its potential impact on diabetes self-management. Guided by this rationale, the study was designed to address the following research questions:

Research Question 1: What types of T2DM self-management content related to the ADCES7 self-care behaviors were accessible in English and Spanish on TikTok?

Research Question 2: Who are the content creators, and what are their differences in video quality scores?

Research Question 3: How do user engagement metrics, such as video length, views, likes, saves, and follower count, differ across T2DM-related TikTok videos?

Methods

Research Design and Data Source

This study employed a cross-sectional descriptive design utilizing TikTok as the primary data source. In January 2024, the first author (LD) queried the platform to identify the top English and Spanish hashtags for T2DM based on the highest video counts. The leading English hashtags ($n = 6$; with video counts in the thousands) were #diabetes (840 000), #diabetic (176 000), #insulinresistance (98 500), #insulin (81 000), #type2diabetes (67 000), and #bloodsugar (52 000). In comparison, the top T2DM Spanish hashtags ($n = 6$) included #diabetes (840 000), #diabetestipo2 (89 500), #insulina (42 900), #glucosa (25 000), #resistenciaainsulina (21 300), and #diabetico (17 000). The #diabetes hashtag is written the same and utilized in English and Spanish videos, hence the large number of videos.

TikTok videos were accessed via a desktop Google Chrome browser (Google, Mountain View, CA, USA). From July 15 to August 30, 2024, a systematic inquiry was conducted utilizing a new TikTok account to ensure that the data collection was not influenced by past research (or personal use).²⁶ During this period, each of the 12 designated hashtags was entered into the search bar to analyze their prevalence and engagement on the platform. Although TikTok's algorithm is proprietary, it is known that videos with higher view counts are promoted to broader audiences and that a higher completion rate (i.e., videos watched in full) further increases visibility.²⁷ To ensure a meaningful representation of substantial viewership, the selected study videos were chosen based on the criterion of having garnered over 500 000 views each. Searches were conducted using TikTok's default "Top" view, which displays videos with the highest view counts, mimicking the typical user experience. The video selection process is shown in Figure 1.

Additional inclusion criteria consisted of (1) English or Spanish language videos, (2) publicly viewable videos, and (3) view counts over 500 000. The first 25 eligible videos per hashtag ($N = 300$) meeting these criteria were included in the study. Videos about type 1 or gestational diabetes, non-English or non-Spanish language videos, those with misspelled or missing hashtags, and duplicate videos were excluded ($n = 385$).

Video Classification Based on ADCES7 Self-Care Behaviors

Videos were classified for the presence of the ADCES7 self-care behaviors: (1) healthy coping, (2) healthy eating, (3) being active, (4) taking medications, (5) monitoring, (6) reducing risk, and (7) problem-solving. The basic metadata on engagement metrics for each eligible video was extracted, including URL, posted date, language, video time length, views, likes, comment count, saves, and follower count.

Content Creator Source Categories

Video sources were categorized into 3 groups: health professionals, personal creators, and companies. In this study, health care professionals on TikTok were identified as content creators who share information about T2DM and portray themselves as possessing expertise in various health care disciplines, including medicine, nursing, and dietetics. The professional credentials were not independently verified. Personal creators are individuals who create content based on personal experiences, opinions, or anecdotal advice, often sharing their diabetes journey or tips without formal health care credentials. A company was defined as any organization or business, such as a health care institution, pharmaceutical company, or wellness brand, that posted content to promote its services, products, or educational information about diabetes.

Video Quality Evaluation

The Global Quality Scale (GQS) is a validated tool designed to evaluate the overall quality, coherence, and usefulness of health information presented in digital online materials. It employs a 5-point scoring scale, ranging from 1

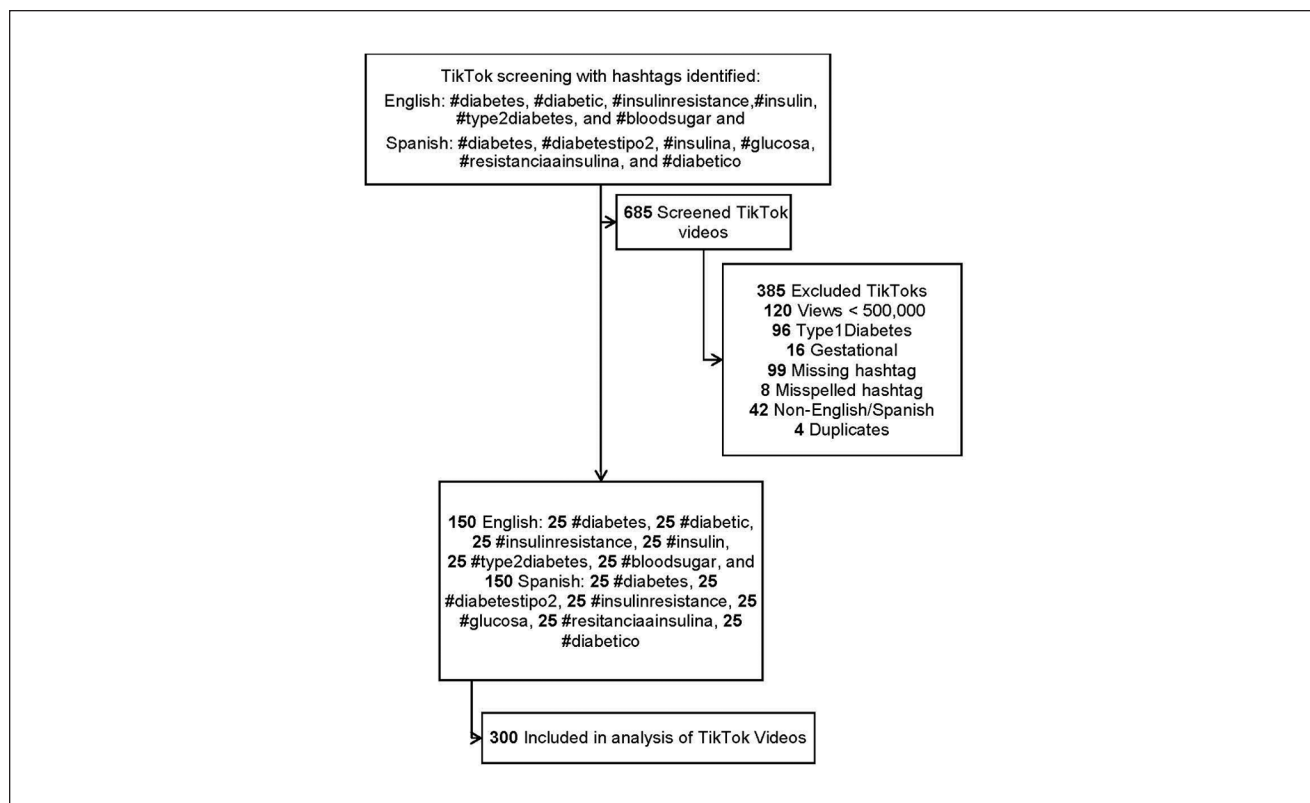


Figure 1. Flowchart for TikTok search query for diabetes videos.

(poor quality, not beneficial for patients) to 5 (excellent quality, highly useful for patients; Table 1).²⁸ Prior studies have utilized the GQS to evaluate health education videos on inflammatory bowel disease, burn care, monoclonal antibody education, and more recently, T2DM content in Spanish and English on YouTube and Facebook.^{13,28-31} Videos with high GQS scores are described as having excellent quality and logical flow and being highly useful for patients,²⁸ suggesting potential to promote engagement, comprehension, and behavior adoption. Given its established validity, the GQS was used in this study to evaluate the quality of TikTok videos about T2DM systematically.

Four independent reviewers (LD, MG, DT, ES) evaluated the videos using the GQS across 2 assessment stages. In the first stage of the analysis, 3 reviewers evaluated the videos. LD, a bilingual certified diabetes care and education specialist (CDCES) with expertise in diabetes clinical care and a nurse researcher experienced in analyzing health content on social media, independently rated all 300 videos. Reviewer MG, a bilingual nurse researcher in Hispanic health, assessed all 150 Spanish-language videos. Meanwhile, reviewer DT, a PhD-prepared nurse specializing in T2DM and digital health, evaluated all 150 English-language videos. During the second stage, a fourth reviewer (ES), a bilingual CDCES with extensive clinical nursing experience in diabetes management, scored all 300 videos. The combined raters' average GQS scores were used to measure video quality characteristics.

Ethical Considerations

An Institutional Review Board application was submitted to the University of California, Los Angeles, Office of the Human Research Protection Program (IRB No. 24-000634) and was determined not to be human subject research because no human participants were involved and all TikTok videos reviewed were publicly accessible. This decision is consistent with determinations for similar descriptive studies involving social media content.^{29,32-34}

Table 1. Global Quality Scale Used to Score Type 2 Diabetes Mellitus Videos on TikTok

Global quality score	Description
1	Poor quality and flow, most information missing; not beneficial for patients
2	Generally poor quality and flow, some information listed, many important topics missing; very limited use to patients
3	Moderate quality, suboptimal flow, some information is discussed but others poorly discussed; somewhat useful for patients
4	Good quality and general flow; most relevant information listed, some topics not covered; useful for patients
5	Excellent quality and flow; highly useful for patients

Only publicly available data were accessed and anonymized where feasible; evaluators adhered to platform policies and user settings and followed the ethical approaches and guidelines for internet research ethics.³⁵

Statistical Analysis

Research Question 1: What types of T2DM self-management content related to the ADCES7 were accessible in English and Spanish on TikTok? Content related to the self-management of T2DM was categorized using the ADCES7 framework to ensure consistency with established DSMES guidelines. A chi-square test was used to assess the associations between language and each of the ADCES7 self-care behaviors. When the expected cell counts were below 5, Fisher's exact test was applied to ensure robust results for small sample sizes.

Research Question 2: Who is the content creator, and what is the quality of T2DM videos on TikTok? For Research Question 2, content creators were categorized into 3 groups: health professionals, personal creators, and companies. Descriptive statistics, specifically, frequencies and percentages, were used to summarize content creator sources. Overall video quality was assessed using the GQS. An independent *t* test was conducted to examine differences in mean GQS scores between English and Spanish video content, and a one-way analysis of variance was performed to evaluate differences in mean GQS scores across the 3 video source categories (health professionals, personal creators, and companies). Post hoc analyses were conducted using Tukey's test for multiple comparisons to further explore significant differences.

Research Question 3: How do user engagement metrics, such as video length, views, likes, saves, and follower count, differ across T2DM-related TikTok videos? User engagement metrics were analyzed, including video duration, views, likes, comments, saves, and follower count, via descriptive statistics, such as counts, means with standard deviations, and medians, to provide an overview of engagement trends. The Kruskal-Wallis H test was used to evaluate the distribution of user engagement metrics. Post hoc pairwise comparisons were performed using Dunn's procedure with Bonferroni correction for multiple comparisons.³⁶ Adjusted *P* values are presented.

Interrater Reliability

Interrater reliability was assessed using weighted Cohen's kappa (Cohen κ) to measure the level of agreement between raters for GQS. The combined raters' GQS scores were used to measure video quality characteristics. The interrater reliability score indicated moderate agreement in the first stage ($\kappa = 0.49$) and substantial agreement between the bilingual CDCES raters in the second stage ($\kappa = 0.71$).

The first author (LD) manually compiled basic information for each video, including Uniform Resource Locator (URLs), and metadata were recorded into Excel (Microsoft Inc.), acknowledging that search results may vary over time due to TikTok's dynamic algorithm and content updates. All analyses were conducted using IBM SPSS Statistics (version 29; Chicago, IL, USA), with the significance threshold set at $P < .05$.

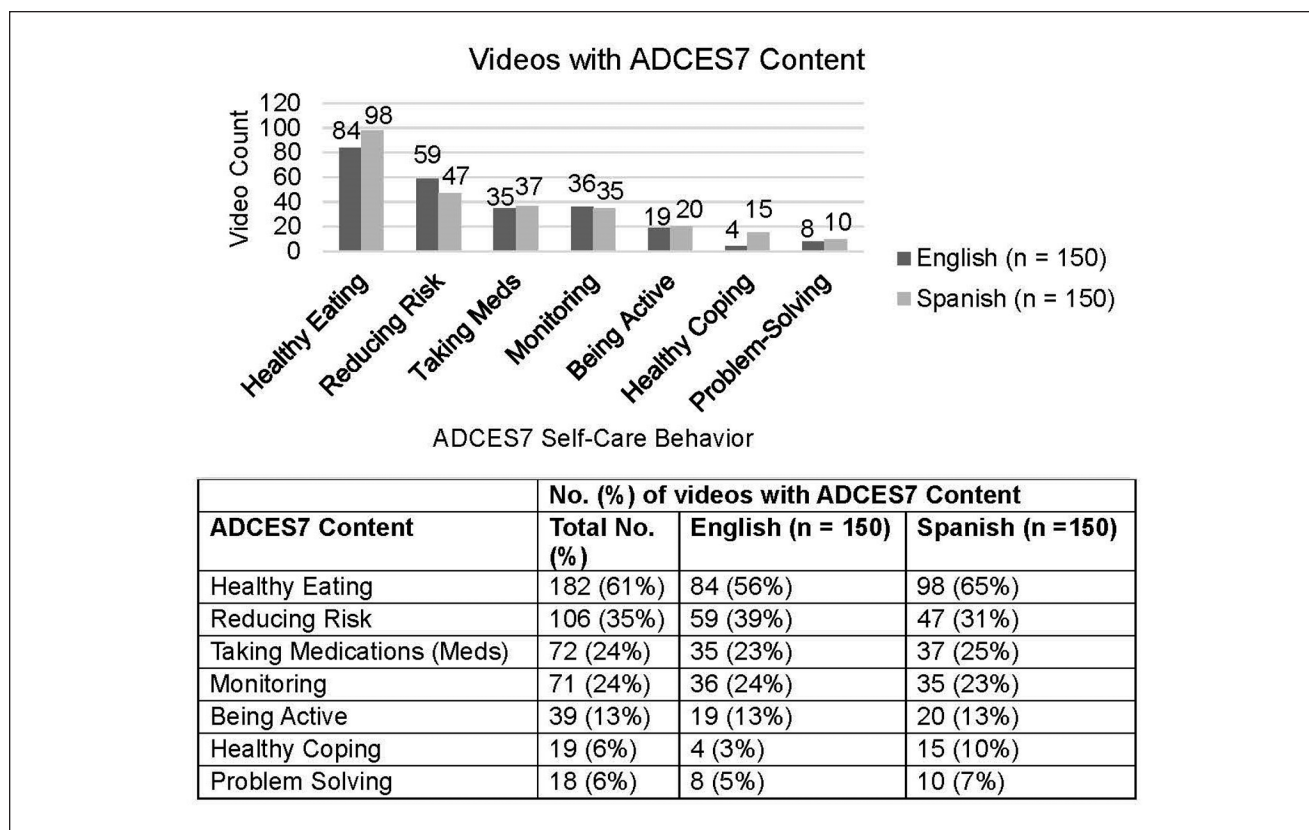


Figure 2. Association of Diabetes Care and Education Specialists 7 self-care behaviors (ADCES) content count by language.

Results

The sample included 300 TikTok videos posted between September 2020 and August 2024, evenly divided between English and Spanish (150 videos per language).

Research Question 1: Language Comparison and Content Classification Using ADCES7 Self-Care Behaviors

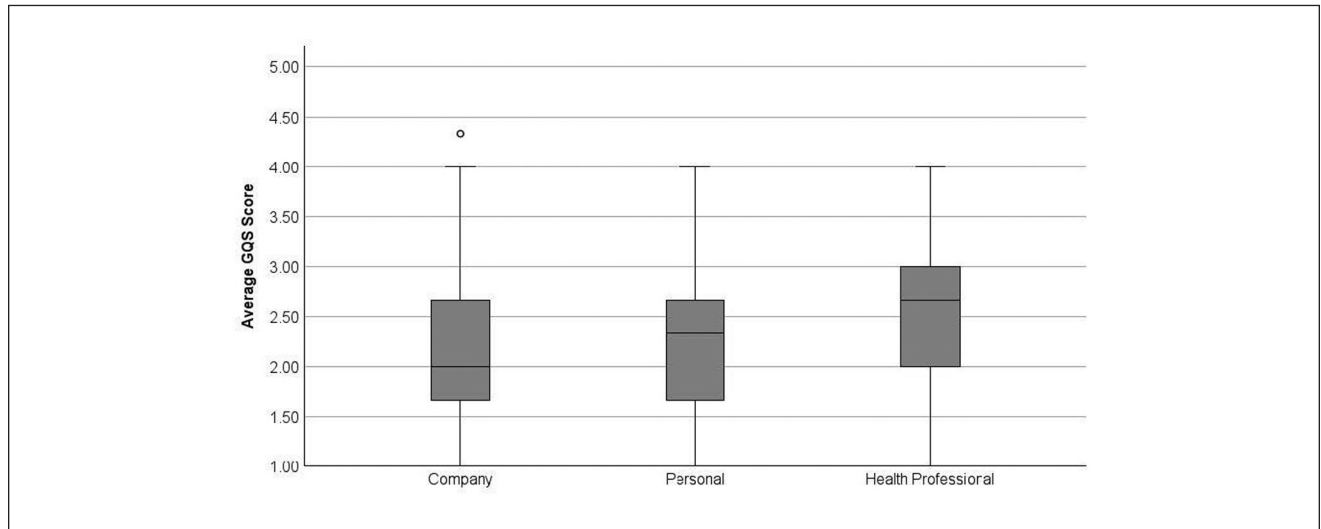
Healthy eating was the most frequently discussed self-care behavior, appearing in 61% of all sampled videos: English: 84 (56%), Spanish: 99 (66%). This was followed by reducing risk, which was mentioned in 35% of all videos: English: 59 (33%), Spanish: 47 (31%). Other self-care behaviors were mainly equally represented across both languages, with no significant statistical differences observed. Problem-solving and “healthy coping” were the least discussed behaviors, appearing in only 6% of the total videos. The term “healthy coping” was featured in Spanish in 10% of Spanish videos (15 videos) compared to only 2% of English videos (4 videos). This difference was statistically significant, $\chi^2_1 = 6.799$, $P < .02$ (two-sided P value for Fisher’s exact test), indicating that healthy coping was more commonly emphasized in Spanish-language content. Figure 2 presents a comparison of the language and content classification of TikTok videos using the ADCES7 framework. Note that several videos addressed multiple topics related to ADCES7, resulting in a total that does not sum to 100%.

Research Question 2: Content Creator and Differences in Video Quality Scores by Language

Regarding video source creators, 35% were created by health care professionals, 34% were from personal accounts, and 31% were from companies, as shown in Table 2.

Table 2. Total Video Count by Language and Content Creator (N = 300)

	Health professional n = 106 (35%)	Personal n = 101 (34%)	Company n = 93 (31%)
English (n = 150)	49 (32%)	64 (43%)	37 (25%)
Spanish (n = 150)	57 (38%)	37 (25%)	56 (37%)

**Figure 3.** Global quality score by content creator.

There was no statistically significant difference in GQS scores between English videos (mean 2.32, SD 0.82) and Spanish videos (mean 2.29, SD 0.87), $t(298) = .410$, $P = .68$. However, GQS scores significantly differed by video content source, $F(2, 297) = 12.93$, $P < .001$. Post hoc Tukey honestly significant difference testing indicated that videos posted by health care professional content creators (mean 2.63, SD 0.80) had significantly higher GQS scores compared to those from personal content creators (mean 2.22, SD 0.82, $P < .001$) and company creators (mean 2.07, SD 0.82, $P < .001$). Conversely, there was no significant difference in GQS scores between the personal and company creator contents. The overall mean GQS score for the TikTok videos was relatively low at 2.31 (2.00–2.67), and a meager 4% of videos scored above 4. As illustrated in Figure 3, the distribution of GQS scores varied across content creators.

Research Question 3: Engagement Metrics Differences Across TikTok Videos

There were no significant differences in video length, views, or saves across the 3 content sources. However, the likes, comments, and follower counts differed significantly according to the content creator (Table 3).

Likes. A Kruskal-Wallis H test indicated a statistically significant difference in the distribution of likes across content creators ($\chi^2 = 9.798$, $P = .007$). Post hoc analysis using Dunn's test with Bonferroni correction showed that company-created videos (median likes, 361 000) received significantly fewer likes than those from personal creators (median likes, 490 000), $P = .005$. However, no significant differences were found between the company and health professionals' videos ($P = .40$) or between health professionals and personal creators ($P = .26$).

Comments. A statistically significant difference was found in the number of comments across content creators. Post hoc analysis revealed that personal creator videos (median: 4817) received significantly more comments than company-created videos (median: 447, $P = .002$). However, no significant differences were found between companies and health professionals ($P = .27$) or between health professionals and personal creators ($P = .25$).

Table 3. Engagement Metrics by Content Creator

Engagement metrics		Time (s)	Views	Likes	Comments	Saves	Followers
Company (n = 93)	Mean	72.52	2 539 255.91	86 646.99	1676.54	23 049.26	58 3141.60
	Mdn	42.00	1 500 000.00	36 100.00	447.00	7443.00	10 5500.00
	SD	89.68	2 977 811.86	225 529.91	4875.43	64 367.12	1 411 063.29
	Minimum	5.00	556 500.00	1757.00	28.00	203.00	3461.00
	Maximum	600.00	20 500 000.00	1 900 000.00	43 000.00	434 800.00	10 700 000.00
Personal (n = 101)	Mean	95.75	2 363 505.94	133 853.54	1881.59	20 368.82	334 353.10
	Mdn	61.00	1 500 000.00	49 000.00	817.00	8617.00	109 300.00
	SD	118.34	2 542 589.83	265 318.11	3097.69	28 207.60	851 375.77
	Minimum	5.00	515 100.00	2815.00	40.00	544.00	346.00
	Maximum	600.00	14 600 000.00	1 900 000.00	19 100.00	160 300.00	5 900 000.00
Health provider (n = 106)	Mean	76.84	2 386 475.47	81 424.82	1107.06	16 748.25	407 828.87
	Mdn	59.50	1 400 000.00	40 600.00	631.00	8305.50	229 100.00
	SD	68.38	2 708 656.86	122 787.94	1485.46	21 586.39	450 591.85
	Minimum	5.00	521 700.00	2600.00	74.00	558.00	1447.00
	Maximum	467.00	17 100 000.00	792 100.00	8805.00	147 900.00	2 200 000.00
Value	Kruskal-Wallis H	3.55	0.80	9.80	11.23	2.55	15.81
	P	.17	.67	.007	.004	.28	<.001

Follower counts. A statistically significant difference in follower counts was observed between personal (the median number was 109 300) and health professional creators (the median number was 229 100, $P < .001$). However, no significant differences were found between the company and personal ($P = .39$) or between health professionals and company creators ($P = .059$).

Discussion

The purpose of this study was to evaluate TikTok videos about T2DM in English and Spanish, focusing on the ADCES7. The types of T2DM self-management content related to ADCES7 self-care behaviors varied substantially across TikTok videos, with few differences observed between content in English and Spanish. Healthy eating was most frequently addressed in both languages, with over 61% of videos featuring this topic. This suggests that TikTok users are highly interested in seeking information on this topic, likely due to its broad appeal and ability to resonate with audiences. Recent research has shown that content related to food and nutrition on social media platforms has gained significant popularity, playing a crucial role in shaping public perceptions and influencing health-related behaviors.³⁷ The widespread presence of such content has prompted health and wellness influencers and self-proclaimed experts to leverage the platforms for engagement, often disseminating misleading or inaccurate information about nutrition.³⁸ Healthy eating and nutrition content dominate social media, giving diabetes educators a strategic role in addressing misinformation. The Academy of Nutrition and Dietetics Standards of Professional Practice and Scope of Practice framework offers clear guidance for identifying and managing food and nutrition topics.³⁸ Adapting this framework for diabetes educators could strengthen their capacity to engage patients and counter misinformation. Future research should evaluate its effectiveness in diabetes education and identify practical adaptations for diverse populations and platforms.

Other ADCES7 self-care behaviors topics appeared less frequently, with problem-solving comprising only 6% of all videos analyzed. This gap represents another opportunity for diabetes educators to address the lacking areas in social media content. To effectively meet people in digital spaces and address current gaps, future content

should offer balanced coverage of all ADCE7 self-care behaviors to support comprehensive diabetes education online. Emphasizing underrepresented ADCE7 self-care behaviors can help individuals navigate daily management challenges, build sustainable self-care habits, and improve overall well-being.

The ADCE7 self-care behavior healthy coping appeared more frequently in Spanish-language videos, indicating this may be a culturally relevant focus on emotional wellness and support among Spanish-language content creators. TikTok's ability to foster shared experiences may help connect Hispanic adults with similar needs, support each other, and seek community and cultural representation. Hispanic adults engage with TikTok more frequently than other ethnic and minority groups,¹⁶ making the platform potentially relevant for promoting healthy coping strategies for T2DM.³ TikTok may increase engagement with these topics by facilitating cultural expression, social interaction, and digital community support. A recent study found that identifying storytellers is crucial for engagement, empowerment, and behavior change in a digital storytelling intervention among predominantly Spanish-speaking Hispanic adults with T2DM.³ TikTok can be an effective community-driven platform, and its influence spans diverse demographics.

This analysis also found that GQS scores were similarly low for both English and Spanish videos, consistent with previous studies that have highlighted the prevalence of low-quality health education on social media. In a study that used GQS to evaluate content on gestational diabetes on TikTok, a mean GQS score of 2.00 was reported.²³ Given the increasing reliance on social media for health information, there is an urgent need to improve the quality of disease-management-related content to ensure users receive accurate and reliable information.

An examination of the GQS score for video content generated by diverse creators has uncovered notable discrepancies in overall quality. The findings indicated that videos originating from personal and corporate sources exhibited markedly lower quality compared to those produced by health professionals. This finding aligns with previous studies that have evaluated health-related video quality.^{22,30,34,39-42} Nevertheless, content from personal accounts tends to attract higher engagement, as reflected in increased likes and comments. It is posited that the preference for personal accounts may stem from their perceived authenticity, genuineness, and relatability among individuals seeking health information. Research indicates that social media users are more inclined to trust personal influencer accounts rather than celebrities, perceiving the former as more reliable and relatable sources of information.⁴³

A number of the low-quality videos examined showed a significant level of viewer engagement, suggesting that viewers may perceive these videos as valuable despite their subpar quality. This discrepancy highlights the risk of misinformation and disinformation on social media, particularly for communities relying heavily on these platforms for health information. The spread of misinformation directly undermines public health efforts, making it essential for health care professionals and public health organizations to engage with and improve health education.⁴⁴ Given these risks, health care professionals should actively engage with patients by asking open-ended questions about their experience with online health-related information, approaching the topic with empathy and nonjudgment, providing evidence-based education where necessary, and improving the accuracy and quality of online health education. Diabetes educators and other health care professionals should be prepared to guide and recommend accurate DSMES social media content to individuals with T2DM. Research findings suggest that professional health organizations, in collaboration with governmental authorities, should strengthen their evaluation and oversight of medical content shared on social media platforms, such as TikTok.²⁵ This is crucial for monitoring online health-related information, future policy development, and enhanced scrutiny of these growing digital spaces. As such, although TikTok may foster community engagement and cultural representation, health care providers should caution users about the limitations and the risk of misinformation on the platform.²⁴

Limitations

The study has several limitations, including a cross-sectional sample of videos from a single social media platform, which may not fully capture the breadth of T2DM-related content. However, TikTok's prominence as one of the fastest-growing social media platforms enhances its relevance, particularly for Hispanic individuals who are increasingly using TikTok.

At the outset of this study, the hashtag #diabetes featured nearly 1 million videos, highlighting the volume of diabetes-related content on TikTok. However, given the constantly evolving nature of TikTok content, this study only captured a brief snapshot of the platform. This study focused on top hashtags related to T2DM, with TikTok also suggesting additional videos upon entering the search hashtag. Additional videos recommended by the TikTok algorithm ($n = 385$) were excluded because they did not meet the inclusion criteria, and therefore, their quality and source were unknown.

Video quality was assessed using the validated GQS tool. However, relying solely on this tool may have limited methodological triangulation; incorporating additional assessment tools such as DISCERN could have strengthened the analysis through a more comprehensive and cross-validated content quality evaluation.²⁴

Although experienced nurses assessed the videos in this study, individuals from other health professions and people with diabetes may interpret content quality differently. Including diverse perspectives could help ensure video content meets the needs of all intended audiences.

Conclusions and Future Implications

Traditional in-person diabetes education has limited reach, creating barriers to DSMES participation, especially for populations with high T2DM prevalence. The findings indicate that TikTok content does not fully address the needs of individuals with T2DM, primarily due to the low video quality and limited coverage of ADCES7 behaviors. Videos from personal accounts had the lowest quality but generated the highest engagement, as measured by likes and comments.

The engagement observed with Spanish-language videos on T2DM highlights the need for researchers and health care professionals to gain a deeper understanding of the TikTok platform, particularly regarding its impact on Spanish-speaking Hispanic adults. Despite these videos demonstrating limited coverage of the ADCES7 self-care behaviors topics and low quality, their significant reach suggests a critical gap in disseminating high-quality educational content. Engaging with prominent content creators can enhance the effectiveness of health communication strategies in promoting diabetes awareness and management within the Hispanic community. Incorporating culturally relevant and comprehensive self-management topics on this platform could address current gaps, enhance engagement, and improve diabetes self-management among this population. TikTok's rapid growth presents an opportunity to expand access to health information more broadly, particularly for users. Social media-based DSMES must prioritize credible, high-quality content from trusted sources to be effective.

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Author Contributions

Conceptualization: Lisa Diaz; Methodology: Lisa Diaz, Dante Anthony Tolentino; Formal analysis and investigation: Lisa Diaz, Dante Anthony Tolentino; Writing - original draft preparation: Lisa Diaz, Dante Anthony Tolentino; Writing - review and editing: Lisa Diaz, Dante Anthony Tolentino, and Sarah E. Choi; Funding acquisition: Lisa Diaz. All authors read and approved the final manuscript.

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Ethics Approval

An institutional review board application was submitted to the University of California, Los Angeles, Office of the Human Research Protection Program (IRB No. 24-000634) and was determined not to be human subject research because no human participants were involved, and all TikTok videos reviewed were publicly accessible.

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Data Availability

The data supporting this study's findings are publicly available via the video-sharing application TikTok.

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